

C:\Users\mthob\OneDrive\Documents\NetBeansProjects\FederatedIDSArtefact\src\main\java\com\registration\app\federatedid

```
1
2 /* Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
3  * Click nbfs://nbhost/SystemFileSystem/Templates/Classes/Class.java to edit this template
4  */
5 package com.registration.app.federatedidsartefact;
6
7 /**
8  * 28/01/2026
9  *
10 * @author mthobi Data loading (CSV--> Memory) - The artefact uses CSV-based
11 * data-sets for transparency and reproducibility
12 */
13
14 import java.io.*;
15
16 import java.util.*;
17
18 public class DataLoader {
19
20     public static List<double[]> loadFeatures(String filePath) throws IOException {
21         List<double[]> features = new ArrayList<>();
22         BufferedReader br = new BufferedReader(new FileReader(filePath));
23         String line;
24
25         // Skip header
26         br.readLine();
27
28         while ((line = br.readLine()) != null)
29         {
30             String[] parts = line.split(",");
31             double[] row = new double[parts.length - 1];
32
33             for (int i = 0; i < parts.length - 1; i++)
34             {
35                 row[i] = Double.parseDouble(parts[i]);
36             }
37             features.add(row);
38         }
39         br.close();
40         return features;
41     }
42
43     public static List<Integer> loadLabels(String filePath) throws IOException {
44         List<Integer> labels = new ArrayList<>();
45         BufferedReader br = new BufferedReader(new FileReader(filePath));
46         String line;
47
48         br.readLine(); // header
49         while ((line = br.readLine()) != null)
50         {
51             String[] parts = line.split(",");
52             labels.add(Integer.parseInt(parts[parts.length - 1]));
53         }
54         br.close();
55         return labels;
56     }
57 }
58
```